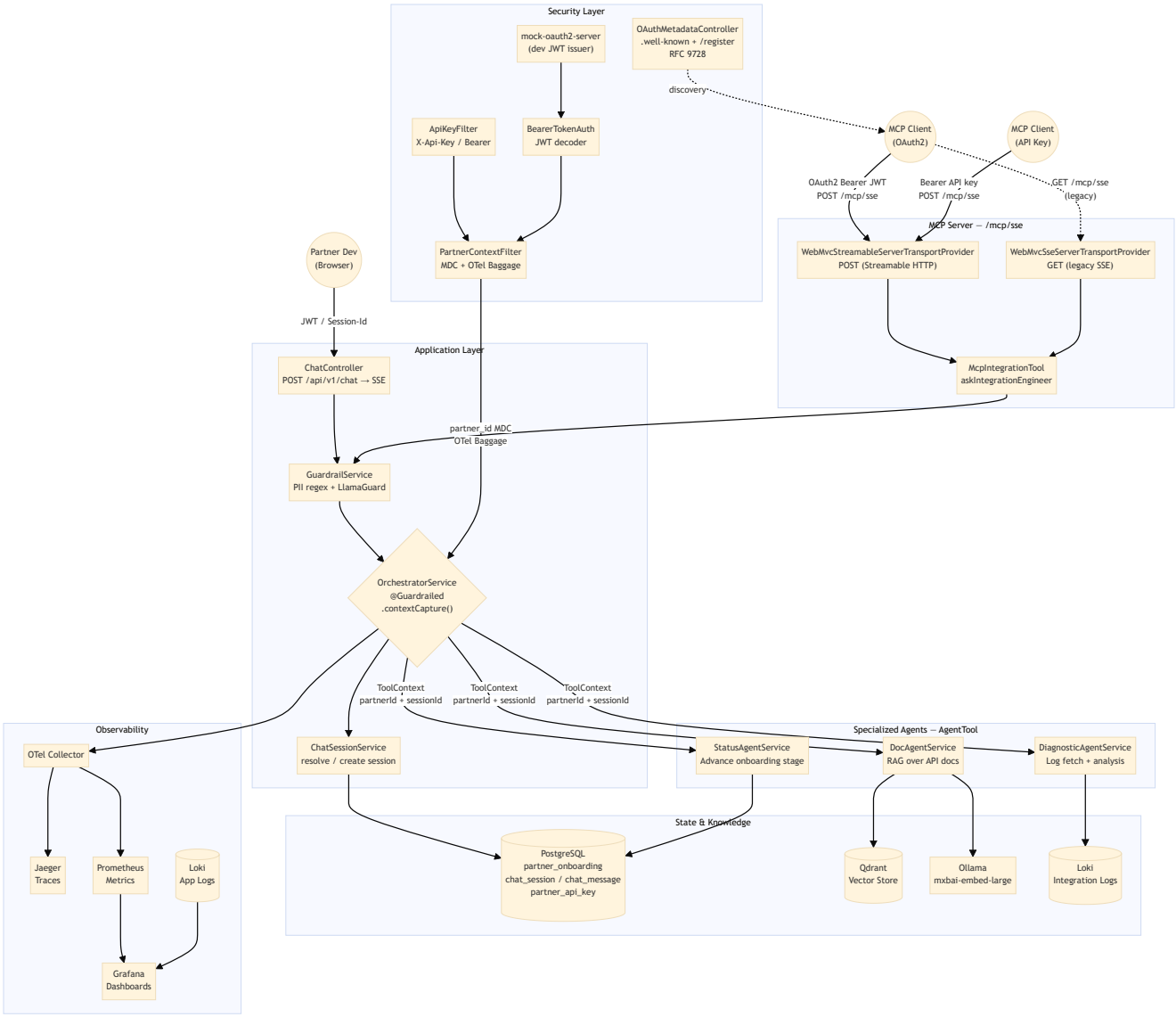
Partner Onboarding Chatbot — Architecture

Overview

A conversational multi-agent system that guides partner engineers through a technical integration roadmap. The chatbot is the interface; behind it, multiple specialized agents coordinate to reason over API documentation, diagnose errors from logs, and advance the partner through each onboarding stage.

Partners can interact through two surfaces: a **web UI** (browser SSE) and an **MCP server** (Claude Code or any MCP-compatible client), both protected by the same authentication and authorization pipeline.

System Design



Authentication & Access

Two authentication methods are supported, processed by the same filter chain:

OAuth2 / JWT

- Partners authenticate via `mock-oauth2-server` (dev) or a real IdP (prod).
- `BearerTokenAuthenticationFilter` decodes the JWT using `NimbusJwtDecoder.withJwkSetUri`, validated against the issuer's JWKS endpoint.
- The JWT `sub` claim becomes the `partnerId`.
- MCP clients discover the OAuth server via `/.well-known/oauth-authorization-server` and `/.well-known/oauth-protected-resource` (RFC 9728). Dynamic client registration is supported at `POST /register`.

API Key

- Partners obtain an API key via `POST /dev/api-key` (hashed with SHA-256, stored in `partner_api_key`).
- `ApiKeyFilter` checks both `X-Api-Key` header and `Authorization: Bearer` header, enabling the same key to work for REST and MCP clients.
- On match, it sets a `UsernamePasswordAuthenticationToken` with the `partnerId` as the principal before `BearerTokenAuthenticationFilter` runs.

`PartnerContextFilter`

Runs after authentication. Extracts `partnerId` from either a `Jwt` principal (OAuth2) or a `String` principal (API key), then:

- Puts `partner_id` in MDC for all log lines on that request.
- Sets `partner.id` as OTel Baggage so it propagates to child spans automatically.

Security configuration highlights

- `RequestAttributeSecurityContextRepository` — saves the `SecurityContext` as a servlet request attribute. Survives Tomcat async dispatches (SSE keepalive, MCP streaming) without re-authenticating.
- `WebSecurityCustomizer.ignoring("/error")` — bypasses the filter chain for Tomcat error-page dispatches so async SSE errors don't cascade into spurious `Access Denied` log noise.
- `@PartnerId` method parameter annotation — `PartnerIdArgumentResolver` extracts `partnerId` from the `SecurityContext` in controller methods, keeping it out of request parameters entirely.

MCP Server

The MCP server exposes the orchestrator as a single tool (`askIntegrationEngineer`) consumable by any MCP-compatible client (Claude Code, Cursor, etc.).

Dual transport at `/mcp/sse`

Transport	Method	Protocol	Used by
`WebMvcSseServerTransportProvider`	`GET`	Legacy SSE	SSE-type MCP clients
`WebMvcStreamableServerTransportProvider`	`POST`	Streamable HTTP	HTTP-type MCP clients (OAuth2 + API key)

Both transports are registered on the same path and coexist via Spring MVC's `RouterFunction` composition. Both use a `contextExtractor` that reads `partner_id` from the JWT subject or String principal and injects it into the `McpTransportContext`.

Session type discrimination

`McpIntegrationTool` detects how the partner authenticated by inspecting the transport context:

- `partner_id` present in transport context → JWT/OAuth2 → `SessionType.MCP_OAUTH`
- Absent (falls back to MDC) → API key → `SessionType.MCP_API_KEY`

This keeps MCP OAuth and API key conversation histories cleanly separated even if the MCP session ID collides.

Orchestrator Design

Streaming response (Reactor Flux)

`OrchestratorService.processInput()` was refactored from a blocking, collect-then-return model to a fully streaming pipeline using Spring AI's `.stream().chatResponse()`:

```
Before: LLM generates full response → collect → return String  (time-to-first-token = full latency)
After:  LLM generates tokens → Flux<String> emitted per chunk   (time-to-first-token ≈ milliseconds)
```

Both consumers handle the `Flux<String>` natively:

- **Web controller** — subscribes and forwards each chunk to a `SseEmitter`, so the browser starts rendering the moment the first token arrives.
- **MCP tool** — collects via `.collectList().map(String::join).block()`. From the MCP caller's perspective this is a single synchronous response — there is no streaming benefit for the end user. Token metering (`doOnNext`) and the latency timer (`doOnComplete`) still fire progressively as tokens arrive from the LLM because those operators sit upstream of `.collectList()` in the Flux chain.

Prompt engineering & token reduction

System prompt and per-stage instructions were rewritten to be directive rather than explanatory. Claude Haiku does not require verbose context to follow structured instructions — concision improves compliance and reduces cost.

Component	Before (est.)	After (measured)	Δ
System prompt	~1,800 tokens	~150 tokens	−1,650

Component	Before (est.)	After (measured)	Δ
Stage instructions	~400 tokens avg	~45 tokens avg	−355
Total per call	**~2,200 tokens**	**~195 tokens**	**−2,005 tokens**

At Haiku input pricing ($0.25 / MTok):

- **Per call:** ~$0.0005 saved
- **At 10k calls/day:** ~$150/month

The reduction also improved **Qdrant retrieval quality**. The original verbose system prompt caused `docAgentTool` to generate broad, context-heavy queries that diluted the embedding similarity score. The tighter prompt produces focused, keyword-dense queries that align better with the chunk embeddings, lifting effective retrieval precision without changing `topK` or `similarityThreshold`.

Agent-to-Agent Architecture

The orchestration model is a **centralized supervisor** — one `OrchestratorService` holds the `ChatClient` and decides which agents to invoke based on the conversation. Agents do not call each other directly.

Routing is LLM-as-router: the model receives all registered tools and decides when to invoke each one based on intent. A partner saying "this isn't working" may trigger both `getLatestLogs` and `docAgentTool` in the same turn.

Agent auto-discovery uses a marker interface:

```
public interface AgentTool {}
```

Any `@Service` implementing `AgentTool` is automatically injected into the orchestrator via `List<AgentTool>`. Adding a new agent requires zero changes to the orchestrator.

Current agents:

Agent	Responsibility
`DocAgentService`	RAG retrieval over API documentation
`DiagnosticAgentService`	Fetch and analyze integration logs via `LogProvider`
`StatusAgentService`	Advance partner through onboarding stages

MDC restoration in agent tools — `RestoreMdcAspect` intercepts all `AgentTool` implementations before each tool invocation: it restores `partner_id` and `session_id` from `ToolContext`, then reads `Span.current()` (the tool child span, made current by `TracingToolCallingManager` before execution) and writes `trace_id` / `span_id` into MDC. Agent tool logs always carry the full trace and partner context with the same `trace_id` as the orchestrator.

Reactor Context Propagation

`OrchestratorService.processInput()` returns a `Flux<String>`. Reactor operators (`doOnNext`, `doOnComplete`, agent tool calls) run on scheduler threads that do not inherit the HTTP request's `ThreadLocal` state. Four complementary mechanisms address this:

1. MDC propagation (partner_id, session_id)

`.contextCapture()` is appended to the Flux chain. At subscription time it snapshots all registered `ThreadLocalAccessor` values — including MDC keys via `Slf4jThreadLocalAccessor` — into Reactor's `ContextView`. `Hooks.enableAutomaticContextPropagation()` (enabled by Spring Boot at startup) causes every downstream operator to restore those MDC thread-locals before executing its callback. `partner_id` (set by `PartnerContextFilter`) and `trace_id` / `span_id` (injected by `processInput()` before the Flux is assembled) are thus available on all scheduler threads without any additional wiring.

2. OTel span propagation into agent tool calls (TracingToolCallingManager)

Spring AI's `DefaultToolCallingManager` executes tool calls synchronously inside a Reactor `flatMap` on a scheduler thread. By the time execution reaches the tool, Micrometer's observation infrastructure has already opened and closed its own OTel scope, clearing `Span.current()`. Without intervention, each tool call produces a disconnected root span with a different trace ID.

`TracingToolCallingManager` wraps `DefaultToolCallingManager` and fixes this: `OrchestratorService` captures `callerSpan = Span.current()` on the request thread before assembling the Flux, places it in `ToolContext` alongside `partnerId` and `sessionId`, and `TracingToolCallingManager.executeToolCalls()` reads it back and calls `callerSpan.makeCurrent()` before delegating. `DefaultToolCallingManager`'s Micrometer observation then creates tool child spans under the now-current parent span. `RestoreMdcAspect` reads the restored `Span.current()` (the child span) and writes `trace_id` / `span_id` into MDC — same trace ID as the orchestrator, different span ID.

The `callerSpan` in `ToolContext` is invisible to `AgentTool` implementations; only the infrastructure layer reads it.

3. Terminal callback (doOnComplete)

`doOnComplete` runs on a terminal signal thread after the Flux completes; the Reactor context restore has already been torn down at that point. `OrchestratorService` reuses the `callerSpan` reference captured in step 2 and calls `callerSpan.makeCurrent()` explicitly inside `doOnComplete` around the token log line so it carries the correct trace.

4. ResponseLoggingFilter

For SSE responses, Spring promotes the connection to async context before the Flux has finished emitting, closing the OTel span scope in the process. `ResponseLoggingFilter` captures the span *before* `chain.doFilter()` while the scope is still open, then injects `trace_id` / `span_id` into MDC around the post-response log line.

Result: every log line — whether from the orchestrator, an agent tool, or the response filter — carries the same `trace_id`, `span_id`, and `partner_id` as the originating HTTP request, regardless of thread switches. In Jaeger, tool call spans appear as children of the orchestrator span within a single unified trace.

State Management

`PartnerOnboarding` — persistent progress record, one per partner, survives across all sessions. Holds current onboarding stage, stage-specific context, and timestamps.

`ChatSession` — one per conversation window. Unique on `(partner_id, session_id, session_type)`. Session type encodes the surface:

session_type	Surface
WEB	Browser UI
MCP_OAUTH	MCP client authenticated via OAuth2 JWT
MCP_API_KEY	MCP client authenticated via API key

The 3-column unique constraint prevents cross-surface session collisions and makes it impossible for a partner to accidentally read another partner's session history. `ChatSessionService.resolveSession` does a direct DB lookup on all three columns — no post-filter needed.

Memory uses a custom `PartnerChatMemoryRepository` backed by PostgreSQL, scoped to `sessionId`. Window is 4 messages (`MessageWindowChatMemory`).

Guardrails & Data Privacy

Tenant isolation is enforced at the tool level via Spring AI's `ToolContext`. The `partnerId` is extracted from the JWT/API key in `PartnerContextFilter` and injected into every tool call. The LLM never supplies or influences the `partnerId` :

```
String partnerId = (String) toolContext.getContext().get("partnerId");
```

PII scrubbing in `GuardrailService.sanitize()` removes sensitive data before it enters or leaves the LLM context window. Patterns covered: credit/debit cards, emails, IPv4, IPv6, Bearer tokens, JWT, API keys (`sk-` , `pk-` , `gsk_`), AWS access keys, phone numbers, CPF, CNPJ.

Content moderation via `GuardrailService` wraps both input and output through a `@Guardrailed` AOP aspect on `processInput`. Production path connects to LlamaGuard via Ollama (`llama-guard3`).

Observability

Distributed tracing — OpenTelemetry with Jaeger. `ObservationConfig` wires a custom pipeline: `ObservationRegistry` → `DefaultTracingObservationHandler` → `OtelTracer` → `OtlpGrpcSpanExporter` → `otel-collector:4317` → Jaeger. Spring Boot's auto-configured OTLP exporter uses HTTP/4318 and does not wire the custom `OtelTracer` ; the manual gRPC configuration is required. `partner.id` and `session.id` are set as OTel Baggage in `PartnerContextFilter` . `TracingToolCallingManager` ensures tool call spans appear as children of the orchestrator span (same trace ID, distinct span IDs) rather than disconnected root spans. Jaeger uses Badger file-based storage (`SPAN_STORAGE_TYPE=badger`) so traces survive container restarts.

Structured logging — Loki via Promtail. Every log line includes `trace_id` , `span_id` , and `partner_id` via MDC. `ResponseLoggingFilter` logs `partner method path → status` for every authenticated MCP and REST request by reading MDC directly (not request attributes). `OpenTelemetryAppender` is intentionally absent from `logback-spring.xml` — it reads `Span.current()` at log time and would overwrite the MDC-propagated `trace_id` with Spring AI's scheduler-thread spans, breaking correlation.

Response headers — `X-Trace-Id` returned on every web API response for support correlation.

Prometheus metrics:

Metric	Tags	Purpose
`orchestrator.llm.latency`	`status`	LLM response time per onboarding stage (histogram)
`doc.agent.calls`	—	Total doc retrieval attempts
`doc.agent.empty.results`	—	Miss rate — gaps in documentation coverage
`diagnostic.agent.calls`	—	Total log fetch attempts
`stage.advancement`	`from` , `to`	Partner progression through stages
`guardrail.blocked`	`direction`	Input/output blocks
`llm.tokens.input`	`partner` , `status`	Input token consumption per partner per stage
`llm.tokens.output`	`partner` , `status`	Output token consumption per partner per stage

`percentiles-histogram: true` on `orchestrator.llm.latency` enables accurate p95 across replicas via `histogram_quantile` .

Grafana dashboards provisioned via `grafana/provisioning/` — survive container restarts, version-controlled alongside code.

Knowledge & Data Access

RAG pipeline:

1. Markdown docs loaded from `docs_folder/` on startup via `ApplicationReadyEvent`
2. Chunked with `TokenTextSplitter` (250 tokens, 100 char minimum)
3. Source filename attached as metadata to every chunk
4. Embedded via `mxbai-embed-large` through Ollama
5. Stored in Qdrant (`initialize-schema: true`)

Retrieval: similarity search with `topK(4)` , `similarityThreshold(0.5)` . Empty results return an explicit message rather than an empty string. Source attribution is included so the LLM can tell partners which document answered their question.

Log retrieval uses a `LogProvider` interface:

- `LokiLogProvider` (`@Profile("!test")`) — queries Loki via LogQL, filters for `/v1/auth` , `/v1/webhooks` , `/v1/orders` , last hour, limit 20
- `MockLogProvider` (`@Profile("test")`) — deterministic per-partner log data for evaluation

Infrastructure

Service	Purpose
PostgreSQL 16	Partner onboarding state, chat sessions, message history, API keys

Service	Purpose
Qdrant	Vector store for RAG
Ollama	Embeddings (`mxbai-embed-large`) + guardrails (`llama-guard3`)
Anthropic Claude	Primary LLM (`claude-haiku-4-5-20251001`)
OTel Collector	Trace and metric collection
Jaeger	Distributed trace visualization
Prometheus	Metrics scraping
Grafana	Dashboards and log correlation
Loki + Promtail	Log aggregation
Mock OAuth2 Server	JWT issuer for local development

Database Schema

```
partner_onboarding           -- one per partner, holds progress
  id UUID PK
  partner_id VARCHAR UNIQUE
  current_status VARCHAR      -- START | AUTH_CONFIGURED | WEBHOOK_SET | LIVE
  context_json JSONB
  created_at, updated_at TIMESTAMP

partner_api_key              -- hashed API keys for direct access
  id UUID PK
  partner_id VARCHAR
  hashed_key VARCHAR UNIQUE   -- SHA-256
  created_at TIMESTAMP
  active BOOLEAN

chat_session                 -- one per conversation window
  id UUID PK
  session_id VARCHAR
  session_type VARCHAR        -- WEB | MCP_OAUTH | MCP_API_KEY
  partner_id VARCHAR FK → partner_onboarding.partner_id
  started_at TIMESTAMP
  UNIQUE (partner_id, session_id, session_type)

chat_message                 -- one per turn
  id UUID PK
  session_id UUID FK → chat_session.id
  role VARCHAR                -- user | assistant | system
  content TEXT
  created_at TIMESTAMP
```

Onboarding Stages

Stage	Mission	Advance condition
`START`	Configure auth, call `/v1/auth`	Auth endpoint returning 200 in logs
`AUTH_CONFIGURED`	Configure webhook endpoint, call `/v1/webhooks`	Webhook endpoint returning 200 in logs
`WEBHOOK_SET`	Validate full flow end to end	Partner confirms events received, logs confirm
`LIVE`	Production — monitor and escalate	Final stage — directs to `#r2-support`

Each stage exposes `getInstructions()` (injected into system prompt) and `getNext()` (used by `StatusAgentService`).

Trade-offs & Future Improvements

Item	Notes
LlamaGuard wiring	`llama-guard3` via Ollama — architecture ready, stubbed due to GPU constraints during development
Rate limiting	`Bucket4j` per `partnerId` — prevent flooding, control LLM cost per partner
MCP client pattern	`LogProvider` interface is the natural MCP client integration point — swap `LokiLogProvider` for a Datadog MCP client when available, zero orchestrator changes
Blue/green doc ingestion	Two Qdrant collections — ingest into shadow, swap on completion — zero-downtime doc updates
Real partner JWT	Replace `mock-oauth2-server` with real identity provider — `SecurityConfig` already uses `NimbusJwtDecoder.withJwkSetUri` for lazy JWKS loading
MCP SSE legacy	`WebMvcSseServerTransportProvider` kept for backward compat — new integrations should use `type: "http"` (Streamable HTTP) in their MCP client config
Kubernetes	Docker Compose for local dev — EKS-compatible with registry and storage class changes only
Async message persistence with encryption	See below

Async chat message persistence with encryption

Current state: `PartnerChatMemoryRepository` writes each `ChatMessage` synchronously to PostgreSQL inside the request thread. This adds DB latency to the streaming critical path and stores message content as plaintext.

Proposed design:

```
OrchestratorService (request thread)
    │  publishes ChatMessageEvent (serialized, encrypted)
```

```
    ▼
Redis Stream  —or—  RabbitMQ Exchange
    │  consumer group (separate thread pool)
    ▼
MessagePersistenceConsumer
    │  decrypts → validates → upserts
    ▼
PostgreSQL  chat_message (content_ciphertext, key_id columns)
```

The `PartnerChatMemoryRepository` read path remains synchronous — it decrypts on fetch so the orchestrator's context window assembly is unaffected. Only the write path becomes async.

Encryption: AES-256-GCM per-partner key, rotatable without re-reading old messages (key ID stored alongside ciphertext). Envelope encryption: data key encrypted with a master key held in HashiCorp Vault or AWS KMS, never in the application config.

Schema additions (Flyway migration):

```sql
ALTER TABLE chat_message
    ADD COLUMN content_ciphertext BYTEA,
    ADD COLUMN key_id             VARCHAR(64),
    ADD COLUMN iv                 BYTEA;          -- GCM nonce
-- content TEXT stays for migration window, then dropped
```

Required dependencies (`build.gradle`):

```gradle
// Async transport — choose one:
implementation 'org.springframework.boot:spring-boot-starter-amqp'         // RabbitMQ
implementation 'org.springframework.boot:spring-boot-starter-data-redis'   // Redis Streams

// Encryption
implementation 'org.springframework.security:spring-security-crypto'       // AES-256-GCM via Encryptors.stronger()

// Key management (production)
implementation 'org.springframework.vault:spring-vault-core:3.1.2'         // HashiCorp Vault transit engine
// or AWS KMS via AWS SDK v2 — no Spring wrapper needed
```

`spring-security-crypto` (`Encryptors.stronger(password, salt)`) provides AES-256-CBC out of the box; for GCM (authenticated encryption, no padding oracle) use `javax.crypto` directly with `AES/GCM/NoPadding` — it is part of the JDK and requires no additional dependency.

Why RabbitMQ over Redis Streams for this use case: RabbitMQ's dead-letter exchange gives automatic retry + poison-message handling without custom consumer logic. Redis Streams require manual `XACK` management and dead-letter wiring. Either works; RabbitMQ is lower operational risk when the payload (plaintext conversation turn) must not be silently dropped.